BT405 Nano biotechnology

Midterm papers spring 2023

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1. **What is BOD? 5 marks**

Biochemical oxygen demand (BOD)

Biochemical oxygen demand is one of the significant parameters used in the estimation of the concentration of biodegradable organic pollutants present in a water sample. In routine practice, BOD determination of any sample is a time consuming process, i.e. 5 days, therefore it is not suitable for online process monitoring.

Its rapid determination of could be possible only by using BOD biosensors. Cells of recombinant Escherichia coli with Vibrio fisheri gene lux AE based biosensor was developed for measuring BOD. Recently, a BOD biosensor has been developed using yeast with oxygen probe and able to detect organic pollutants within 15 minutes.

2. **What are biosensors elements? 3 marks**

Biosensor elements:

-A variety of substances may be used as the bio element in a biosensor. Examples of these include:

• Nucleic acids

• Proteins including enzymes and antibodies. Antibody-based biosensors are also called
immunosensors.

• Plant proteins or lectins

• Complex materials like tissue slices, microorganisms and organelles

3. **Basic Characteristics of a Biosensors? 3 marks**

1. LINEARITY: Linearity of the sensor should be high for the detection of high substrate concentration.
2. SENSITIVITY: Value of the electrode response per substrate concentration.
3. SELECTIVITY: Chemicals Interference must be minimized for obtaining the correct result.
4. RESPONSE TIME: Time necessary for having 95% of the response

4. **Difference between properties of nanotubes of Electrical or thermal properties? 5 marks**

Electrical Properties:
If the nanotube structure is armchair then the electrical properties are metallic. If the nanotube structure is chiral then the electrical properties can be either semiconducting with a very small band gap, otherwise the nanotube is a moderate semiconductor. In theory, metallic nanotubes can carry an electrical current density of 4×109 A/cm2 which is more than 1,000 times greater than metals such as copper.

Thermal Properties:
All nanotubes are expected to be very good thermal conductors along the tube, but good insulators laterally to the tube axis. It is predicted that carbon nanotubes will be able to transmit up to 6000 watts per meter per Kelvin at room temperature; compare this to copper, a metal well-known for its good thermal conductivity, which transmits 385 watts per meter per K. The temperature stability of carbon nanotubes is estimated to be up to 2800oC in vacuum and about 750oC in air.

5. **Describe Carbon Nano tube properties**? **5 marks**

Carbon Nanotubes (CNT) Properties:

- CNTs have High Electrical Conductivity
- CNTs have Very High Tensile Strength
- CNTs are Highly Flexible- can be bent considerably without damage
- CNTs are Very Elastic ~18% elongation to failure
- CNTs have High Thermal Conductivity
- CNTs have a Low Thermal Expansion Coefficient
- CNTs are Good Electron Field Emitters
- CNTs Aspect Ratio

6. **Define porous alumina, zeolites and aerogels? 5 marks**

Porous alumina:

Porous alumina membranes produced by anodic aluminum oxidation are characterized by having hexagonally close-packed channels with diameters ranging from 10 to 250 nm or greater. This material is often used as a template for the synthesis of other materials.

Zeolites:

Zeolites are natural crystalline materials with pores having regular arrangements. They are also often used in the template synthesis of nanomaterial. Selenium can be incorporated into the channels of mordenite - a natural zeolite

Aerogels:

Silicon dioxide (SiO_2) is the main component of quartz. It is chemically robust and finds widespread applications. Aerogels are manufactured with a sol-gel technique and can be made of carbon, metal oxide, polymers or silicates.

7. **Describe Liquid crystal applications? 5 marks**

The order of liquid crystals can be manipulated with mechanical, magnetic or electric forces. What is interesting is that this change of

order can be obtained with very small variations of these forces. The properties of liquid crystals are useful in many applications.

The color of some liquid crystals depends on the orientation of its molecules, so any influence that disturbs this orientation (e.g. a difference in magnetic or electric field, temperature, or the presence of certain chemicals) can be detected with a color change.

Liquid crystals are routinely used in displays for cell phones, cameras, laptop computers and other electronics. In these displays, an electric field changes the orientation of the molecules in the liquid crystal, and affects the polarization of light passing through them.

Because of their sensitivity to temperature, and the property of changing color, they are also used in thermometers. In miniaturized sensors, liquid crystals can detect certain chemicals, electric fields and changes in temperature.

8. **Explain Different Liquid Crystal phase? 3 Marks**

Liquid crystals Phase:

A fluid phase in which a liquid crystal flows and will take the shape of its container. It differs from liquid that there is still some orientation order possessed by the molecules**.**

A phase that exists between solid and liquid. Discovered in 19th century when studying a cholesterol derivative.

- **Nematic**
- **Sematic**
- **Cholesteric**

Nematic Phase: Molecules in this phase are long and rod-like in shape. They are free to move in space.

Sematic Phase: This phase can be reached at lower temperatures than the Nematic phase. Molecules align themselves in layers. More order and higher viscosity

Cholesteric Liquid Crystal: Molecules with intermolecular forces that favor alignment between molecules at a slight
angle to one another. The director is not fixed in space as in a Nematic phase, it rotates throughout the sample

9. **Define Nano porous, Nano crystalline, Nano composite and hybrid material? 5 marks**

Nanoporous materials have nano-sized pores.

Nanocrystalline material consists of many nano-sized crystalline domains. One example is nanocrystalline materials, which are polycrystalline (i.e. made of many crystals which are
identical but connected without orientation) and defined as materials with grain sizes from a few nanometers up to 100 nm. In contrast, the grain size in industrial metallic materials is about 10 000 nm or greater. These materials generally show improved mechanical properties (toughness, hardness, etc.).
Nanocomposite material contains two or more phase separated components with morphology of spheres,
cylinders or networks with nano-sized dimensions (further divided into inorganic and polymer nanocomposites).

Hybrid materials are made of a combination of organic and inorganic components interconnected at a molecular
level (e.g. block copolymers)

10. **Define Quantum confinement 3 marks**

Quantum confinement in a nanomaterial, such as a metal, electrons are confined in space rather than free to move in the bulk of the material.

11. Magnetic Resonance Properties of AuNPs for MRI 5 marks

The new era of molecular imaging, in vivo characterization, and measurement of biological processes at cellular and molecular level aims at quantifying molecular changes associated with the onset and development of pathological states, thereby, providing early diagnosis and prognosis of diseases like cancer.

The imaging of cells, cellular and subcellular structures, requires imaging agents of high relaxivity and density endowed with targeting ability to specific cellular receptors.

AuNPs are especially attractive for imaging and therapy due to their SPR, enhanced light scattering, and absorption. In the field of MRI, AuNPs can be used as a template agent in place of Gadolinium (Gd) chelates for use as MRI contrast agents due to their high sensitivity and also in diagnosis.
PEG-coated iron oxide gold is a type of core-shell nanoparticles (PEG-AuIONs), measuring approximately 25 nm in diameter, which reveals high specificity to solid tumors as they accumulate mostly within the tumor mass along with low nonspecific accumulation in the liver and spleen. PEG-AuIONs as promising MRI contrast agents for diagnosis of malignant tumors including pancreatic cancer.

12. Applications of Nanotechnology in Diagnostic? 5 marks

Several nanoparticles have been used for diagnostics. Of these, the most frequently used are gold
nanoparticles, QDs, and magnetic nanoparticles. Nano diagnostics is the new term that describes use of
methods and techniques of nanotechnology and its principles for diagnostics purposes. It includes, although not limited to, the manipulation and assessment of single molecule, size reduction of

systems and platforms to make use of nanoscale properties obtainable from interactions between surfaces and biomolecules.

Nano diagnostics is an evolving application of nanoscale technology to meet the demand of clinical diagnostics, determining disease state, any predisposition to such, the pathology of the condition and the identification of causative organisms.

With nanotechnology, diagnosis is being carried out on a nano-scale leading to a trend of the use of hand held devices that are easy to use and marketable.
Nano diagnostics as surging new field of molecular diagnostics, have been positively changing laboratory procedures, providing new ways for patient's sample assessment and early detection of disease biomarkers with increased sensitivity and specificity.

Nanoparticle platforms have been developed and optimized for the detection of pathogens and cancer biomarkers such that diagnostic procedures now become less cumbersome but more sensitive because most of the complex procedures are now integrated onto a simple device having the capacity to be used for on the spot diagnosis.

13. Uses of Nano biotechnology? 5 marks

With the use of nanotechnology, scientists hope to prevent illness, more quickly diagnose, control disease and
treat disease with fewer side effects, and create better medical aids such as more compatible prosthetics.
Nanoparticles and surfaces made of nanostructures are used in many areas of healthcare research. Specific
applications for nanotechnology in medicine include these developments:
- Better tools for prevention

- Nano scopes and Nano tweezers
- Novel membranes for cleaning blood
- Miniaturized probes for recognizing disease

➢ Nano-dots that trace disease

➢ Improved detection through medical imaging

- MRI (Magnetic Resonance Imaging) with better contrast agents
- CAT (Computerized Axial Tomography) scans

➢ Innovative medicines and drug delivery for detection and treatment

14. How nanotechnology influence the Environment? 5 marks

Safer and more efficient approaches to waste management, air and water purification, reduction in pollution.

Nano filtration to remove dissolved salts, heavy metal contaminants, soften water and treat wastewater.

Nanofibers filtrate dangerous bacteria in the air, gold nanoparticles clear volatile organic compounds.

Developed by the oil industry, MCM-41 (known also as "self-assembled monolayers on mesoporous supports," SAMMS), with pore sizes in the range of 10—100 nanometers, is used for the removal of ultrafine contaminants.

A nanoparticle-reinforced polymeric material can replace structural metallic components in automobiles and lead to a reduction of 1.5 billion liters of gasoline consumption over the life of one year's production of vehicles, thereby reducing carbon dioxide emissions annually by more than 5 billion kilogram.

15. Role of nanotechnology in medicines 5 marks

Nanotechnology in Medicine:

Nanotechnology is a new field with many possible uses, medicine being one of them

❖ The manufacturing technology of the 21st century

The study and manufacture of devices of molecular dimensions, in the range of nanometers or one billionth of a meter. Most of industrial manufacturing processes are based on top-down technologies i.e., they take larger objects and make them smaller yielding products of fairly high precision and complexity. Most products of living organisms are constructed by tiny molecular machines, such as cells and organelles, working from the bottom up. By organizing individual atoms and molecules into particular configurations, these molecular machines are able to create works of astonishing complexity and size, such as the human being Nature shows that molecules can serve as machines because living things work by means of such machinery.

Enzymes are molecular machines that make, break, and rearrange the bonds holding other molecules together. Muscles are driven by molecular machines that haul fibers past one another.

DNA serves as a data-storage system, transmitting digital instructions to molecular machines e.g., the ribosomes, that manufacture protein molecules. Using special bacterium-sized "assembler" devices, nanotechnology would permit on a programmable basis exact control of molecular structures that are not readily manipulated by natural molecular machines and molecular techniques presently available.

With nanotechnology, atoms will be specifically placed and connected, all at very rapid rates, in a fashion similar to processes found in living organisms

Nano medicine:
Some medicines are made through biotechnological processes, for example those using recombinant DNA (human hepatitis vaccine). Under these processes the DNA of living creatures (usually bacteria) is altered. Nanotechnology represents a similar approach to the manufacture of pharmaceuticals and other goods.

16. **Applications of Nanotechnology? 5 marks**

Nanotechnology is the act of manipulating materials at very tiny scales – at the level of atoms and molecules
Two principal parts to defining what is to be considered **nanotechnology:**
(i) Scale and (ii) Uniqueness/novelty
Nanotechnology is the understanding and control of matte
(i) at dimensions between approximately 1 nm to 100 nm
(ii) where unique phenomena enable novel applications

- Nanotechnology is one of the leading scientific fields today since it combines knowledge from the fields of Physics, Chemistry, Biology, Medicine, Informatics, and Engineering. It is an emerging technological field with great potential to lead in great breakthroughs that can be applied in real life.
- Novel Nano and biomaterials, and Nano devices are fabricated and controlled by nanotechnology tools and techniques, which investigate and tune the properties, responses, and functions of living and non-living matter, at sizes below 100 nm.
- The application and use of nanomaterials in electronic and mechanical devices, in optical and magnetic components, quantum computing, tissue engineering, and other

biotechnologies, with smallest features, widths well below 100 nm, are the economically most important parts of the nanotechnology nowadays and presumably in the near future.

- The number of Nano products is rapidly growing since more and more Nano engineered materials are reaching the global market. The continuous revolution in nanotechnology will result in the fabrication of nanomaterials with properties and functionalities which are going to have positive changes in the lives of our citizens, be it in health, environment, electronics or any other field.
- In the energy generation challenge where the conventional fuel resources cannot remain the dominant energy source, taking into account the increasing consumption demand and the CO2 emissions alternative renewable energy sources based on new technologies have to be promoted.
- Innovative solar cell technologies that utilize nanostructured materials and composite systems such as organic photovoltaics offer great technological potential due to their attractive properties such as the potential of large-scale and low-cost roll-to-roll manufacturing processes. The advances in nanomaterials necessitate parallel progress of the Nano metrology tools and techniques to characterize and manipulate nanostructures.
- Revolutionary new approaches in Nano metrology will be required in the near future and the existing ones will have to be improved in terms of better resolution and sensitivity for elements and molecular species. For centuries, silver has been used for its ability to destroy bacteria — from ancient Romans treating their water with silver coins to NASA using the metal to purify water aboard the Space Shuttle.
- **Silver(Ag) nanoparticles** are embedded in sticking plasters for their ability to inhibit the transmission of viruses.

- **Pancreatic cancer** has a devastatingly low survival rate (less than 5 percent after 5 years) because it is usually diagnosed at an advanced stage. Scientists have created tools for the early diagnosis of pancreatic cancer by attaching a molecule that binds specifically to pancreatic cancer cells to iron oxide nanoparticles that are clearly visible under magnetic resonance imaging (MRI).
- If you hate injections, you'll be glad to hear that oral administration of drugs that are currently delivered by injection may be possible in many cases. The drug is encapsulated in a nanoparticle which helps it pass through the stomach to deliver the drug into the bloodstream.
- There are efforts underway to develop oral administration of several different drugs using a variety of nanoparticles. One company has progressed to the clinical testing stage with a drug for treating systemic fungal diseases.
- **Titanium dioxide** confers the white appearance of high-protection sunscreens. Titanium oxide nanoparticles have a comparable UV protection property to the bulk material, but lose the cosmetically undesirable whitening since the particle size is decreased.
- Nanotechnology can be used to create fabrics with superior performance without compromising the look, feel or comfort of the fabric. For instance, nanomaterials can be added to the fabric and make them stain resistant

17. Self-Assembled monolayers (SAMs)? 3 marks

Some organic molecules, when exposed from solution or vapor to a suitable substrate, self-assemble to produce homogeneous, densely packed layers of monomolecular thickness. These organic molecules have long chains with two different end groups. The monolayer is

formed when one of the two end groups of the organic molecule reacts with a particular surface forming a chemical bond. The surface properties of the substrate are then defined by the exposed functional groups of the monolayer. For example, alkyl-saline or alkane thiol molecules exposed to a silica or metal surface assemble into organized layers.

18. **Inorganic nanocomposites? 3 marks**

High-performance ceramics are sought in many applications, such as highly efficient gas turbines, aerospace materials, cars, etc. The field of ceramics that focuses on improving their mechanical properties is referred to as structural ceramics.

Nanocomposite technology is also applicable to functional ceramics such as ferroelectric, piezoelectric, varistor and ion-conducting materials. In this case, the properties of these nanocomposites relate to the dynamic behavior of ionic and electronic species in electro-ceramic materials. Among these materials, in this document, the review is limited to nanocomposite with enhanced magnetic properties.

19. **Nano coating Applications? 5 marks**

Current applications for its technology are in the categories of Electronics, Lifestyle, Life Sciences, Filtration & Energy and Military & Institutional. Important area of application of Nano coatings is tribological coatings.

Tribology is the science and technology of interacting surfaces in relative motion. Tribological properties include friction, lubrication and wear. Tribological coatings are those coatings that are applied to the surface of a component in order to control its friction and wear. In this area, the term ‘thin films’ is often used as an alternative to Nano coatings. Traditional materials used in coatings for tribological applications are carbides, cemented carbides, metal ceramic oxides, nitrides and carbon-based coatings

20. Difference between super hydrophilic coatings and super hydrophobic coatings? 5 marks

Superhydrophilc coatings:

Photocatalytic coatings (commercialized as self-cleaning glass) which use the catalytic properties of titanium dioxide (TiO2). When irradiated with UV light, the coating becomes superhydrophilc: therefore, rain water adheres to the glass providing 'selfcleaning'. Pilkington Activ™ Self-cleaning Glass is a commercial example of a glass with a photocatalytic coating that renders the material easier to clean.

Superhydrophilc coatings are also useful for rendering a surface fog-resistant. Substances that can be used as anti-fog agents include surfactants (e.g. soap), hydrogels, hydrophilic nanoparticles and colloids. The anti-fog agent creates a thin film that does not allow the formation of water droplets but rather 'forces' the water molecules to spread on the surface.

Superhydrophobic coatings:

Nano coating is hydrophobic (water repellent), oleo phobic (oil repellent) surface layer that repels water, oil, dirt, and other dry particles. High hydrophobicity eco-friendly nano coating can be applied onto objects to make them waterproof for the long-term. The opposite to superhydrophilc coatings are super hydrophobic coatings, which totally repel water. Droplets of water on these surfaces have very high contact angles and 'bead-up' forming nearly spherical droplets. Super hydrophobic coatings have surfaces that mimic the surface found in the lotus leaf and are being developed for many applications that require resistance to dirt and ease of cleaning.

21. What is optical microscope and its resolving power? 3 marks

An optical microscope uses visible light (i.e. electromagnetic radiation) and a system of lenses to magnify images of small samples. For this reason, it is also called a light microscope. Optical microscopes are the oldest and simplest of the microscopes.
The resolution limit of an optical microscope is governed by the wavelength of visible light. Visible light is the part of the electromagnetic spectrum with wavelengths between 400 and 700 nm. The resolving power of an optical microscope is around 0.2 µm or 200 nm: thus, for two objects to be distinguishable, they need to be separated by at least 200 nm.

22. Describes the Types of optical sensors? 3 marks

Types of Optical Sensors:
There are different kinds of optical sensors, the most common types which we have been using in our real world applications as given below. Photoconductive devices used to measure the resistance by converting a change of incident light into a change of resistance. The photovoltaic cell (solar cell) converts an amount of incident light into an output voltage. The Photodiodes convert an amount of incident light into an output current.

- **Through-Beam Sensors:**
  The system consists of two separate components the transmitter and the receiver are placed opposite to each other. The transmitter projects a light beam onto the receiver. An interruption of the light beam is interpreted as a switch signal by the receiver. It is irrelevant where the interruption occurs.
- **Retro-Reflective Sensors:**
  Transmitter and receiver are both in the same house, through a reflector the emitted light beam is directed back to the receiver. An

interruption of the light beam initiates a switching operation. Where the interruption occurs is of no importance.

- **Diffuse Reflection Sensors:**
  Both transmitter and receiver are in one housing. The transmitted light is reflected by the object to be detected.

23. Write note on biosensor type of Transducer? 5 marks

The transducer is an element that converts one form of energy into another.

In a biosensor the role of the transducer is to convert the bio-recognition event into a measurable signal.

This process of energy conversion is known as signalization.

Most transducers produce either optical or electrical signals that are usually proportional to the amount of analyte–bioreceptor interactions.

24. Write note on calorimetric sensors? 5 marks

Calorimetric sensors (Note the spelling; Colorimetric sensors will refer to optical sensors!!), involve the measurement of heat that is generated (Calorie: unit of heat; Calorimetric: Measurement of heat). These sensors typically utilize thermistors that transform heat generated or loss during a reaction into an electrical signal.

The entire sensing set-up is surrounded by an insulated jacket to ensure no loss of heat generated during the reaction. The enzyme is immobilized in a packed bed column that is placed in the center of the insulated chamber. The sample is allowed to pass through a coiled column of aluminum that serves as a heat exchanger to ensure uniform temperature of the sample entering the reaction chamber.

A thermistor is placed at the inlet of the reactor containing the immobilized enzyme and another thermistor is placed at the exit of the reactor to measure the temperature after the conversion of the substrate to product.

A thermistor records temperature changes by altering its resistance.

Higher temperatures will causereduction in the resistance of the thermistor. The difference in resistance between thermistors located at the entry and exit points to the reactor is recorded as a measure of the temperature change produced as a result of the passage of the sample.